

Assignment-2

Applying Microservices & Event-Driven Architecture in E-Commerce

In a hypothetical e-commerce platform, Microservices and Event-Driven Architecture are implemented to handle complex operations like order processing, inventory management, and user notifications efficiently.

Microservices Architecture

- Each feature is broken into small, independent services: Order Service, Inventory Service, Payment Service, Notification Service.
- Services communicate using REST APIs, enabling flexibility and independent deployment.
- Ensures scalability and fault isolation.

Event-Driven Architecture

- Services communicate asynchronously through events.
- Example: When an order is placed, the Order Service emits an 'OrderPlaced' event.
- Inventory Service listens to 'OrderPlaced' to update stock.
- Notification Service listens to send email/SMS to the customer.

How SOLID Principles Enhance Design

- S (Single Responsibility): Each microservice handles one specific task.
- O (Open/Closed): Services can be extended for new features without modifying existing code.
- L (Liskov Substitution): Interfaces allow services to be replaced with minimal change.
- I (Interface Segregation): Services expose only relevant endpoints.
- D (Dependency Inversion): High-level modules depend on abstractions (event contracts, APIs).

Observing DRY and KISS

- DRY (Don't Repeat Yourself):
 - Common logic is placed in shared libraries (e.g., authentication, logging).
 - Reusable event formats across services.
- KISS (Keep It Simple, Stupid):
 - Each service performs only necessary tasks.
 - Clear communication through simple event structures.

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